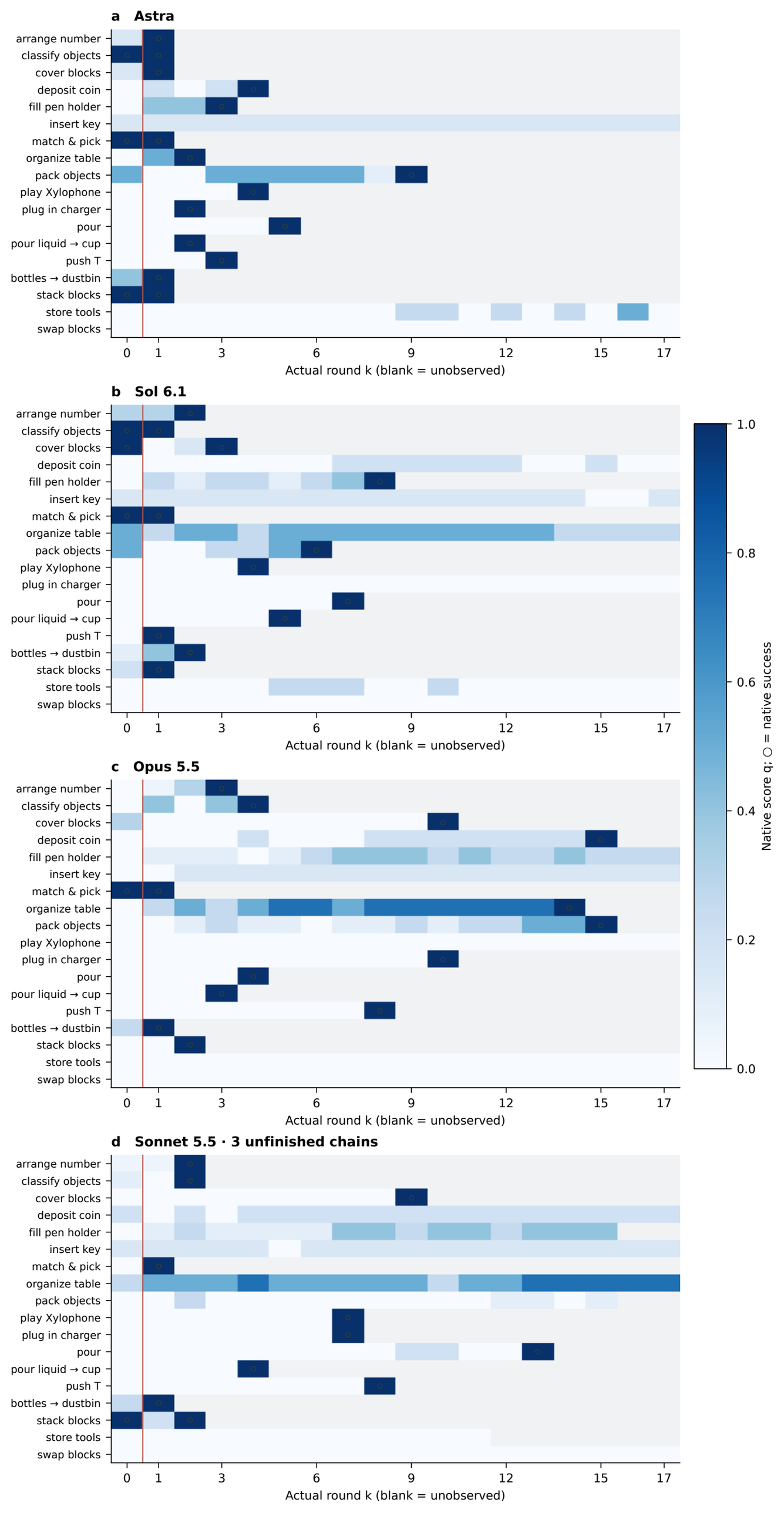
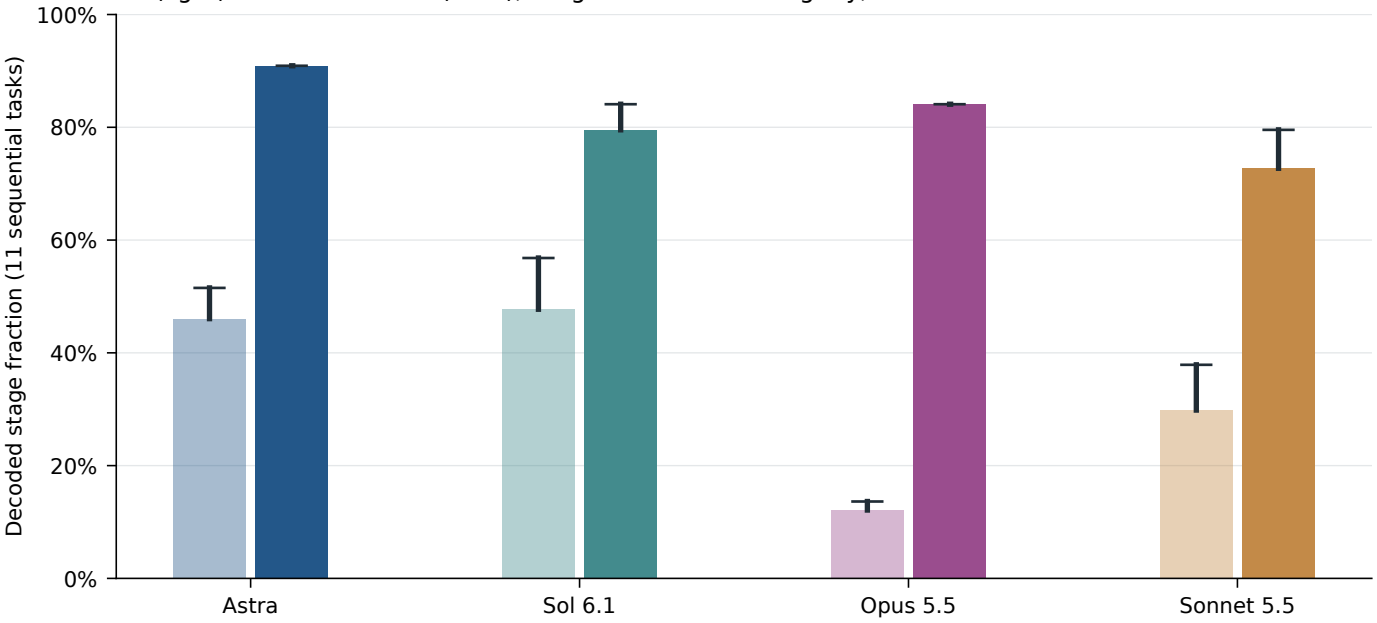
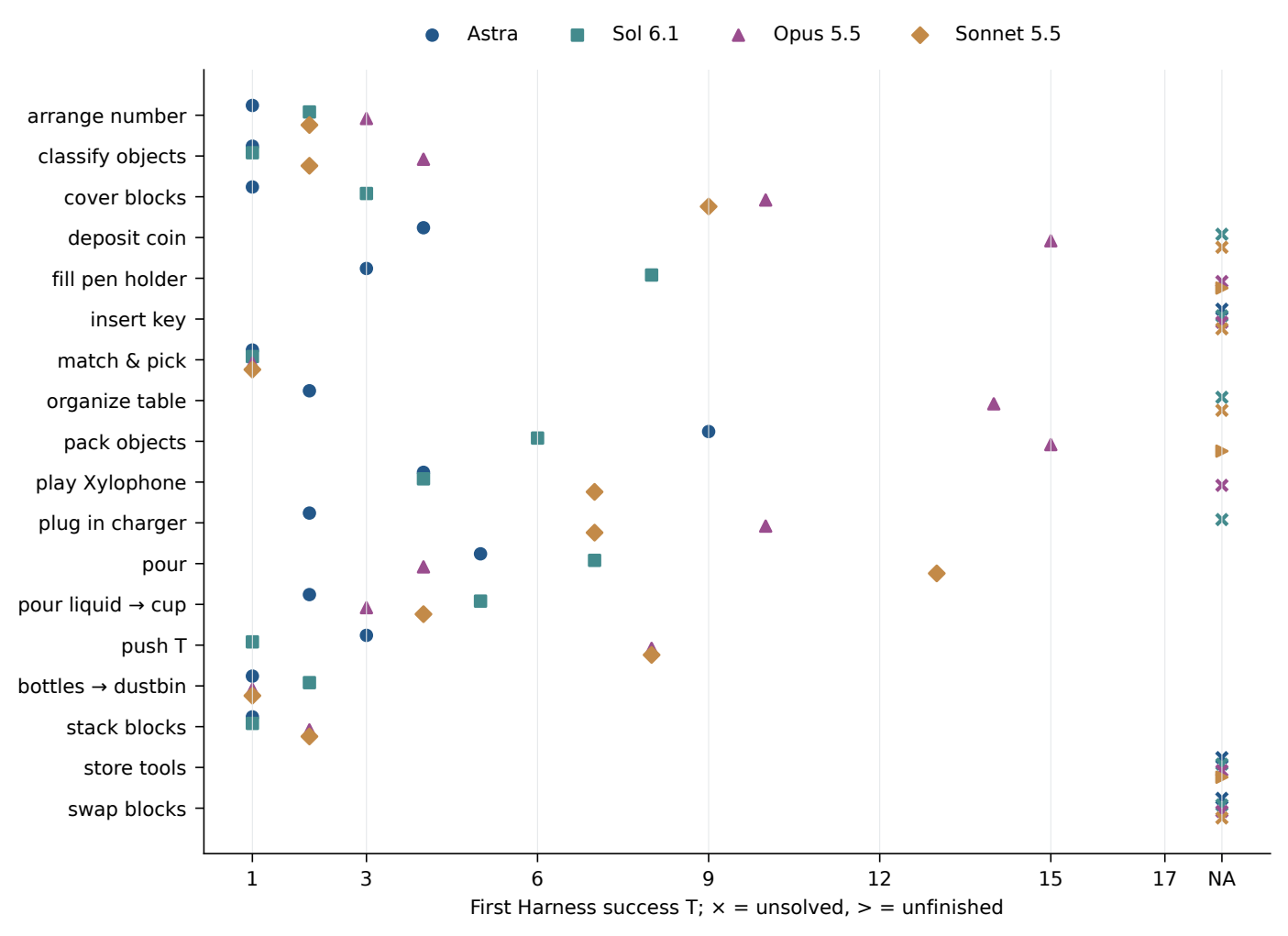
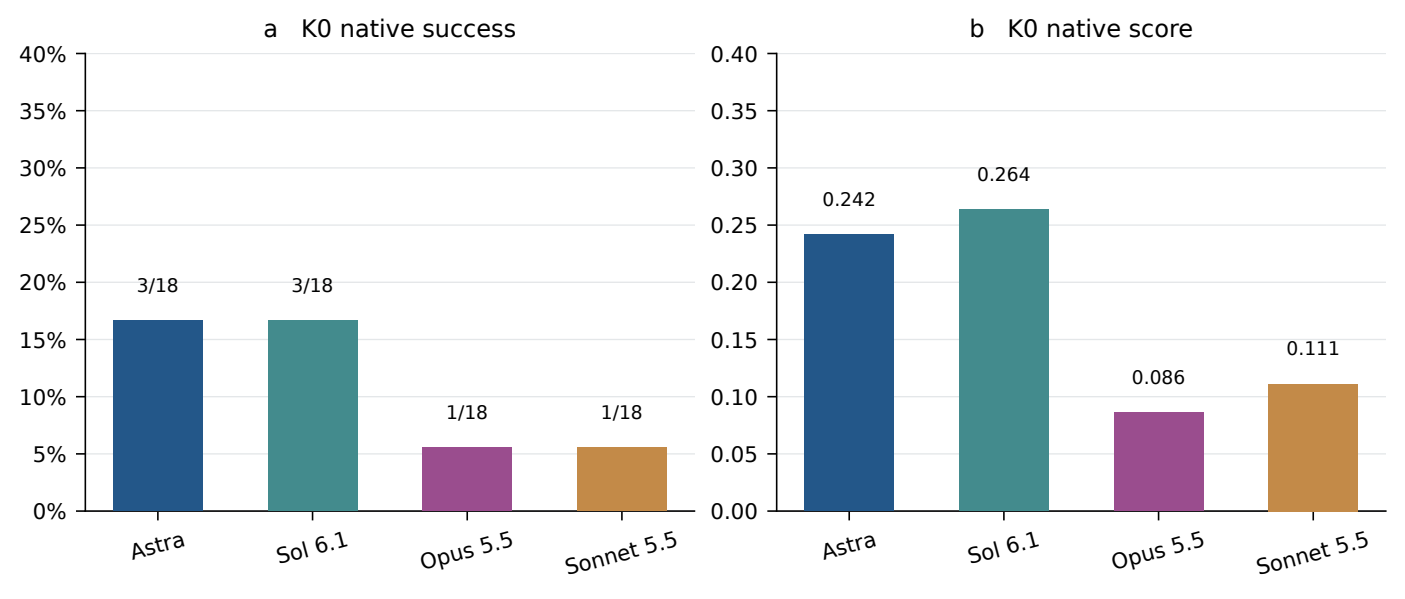
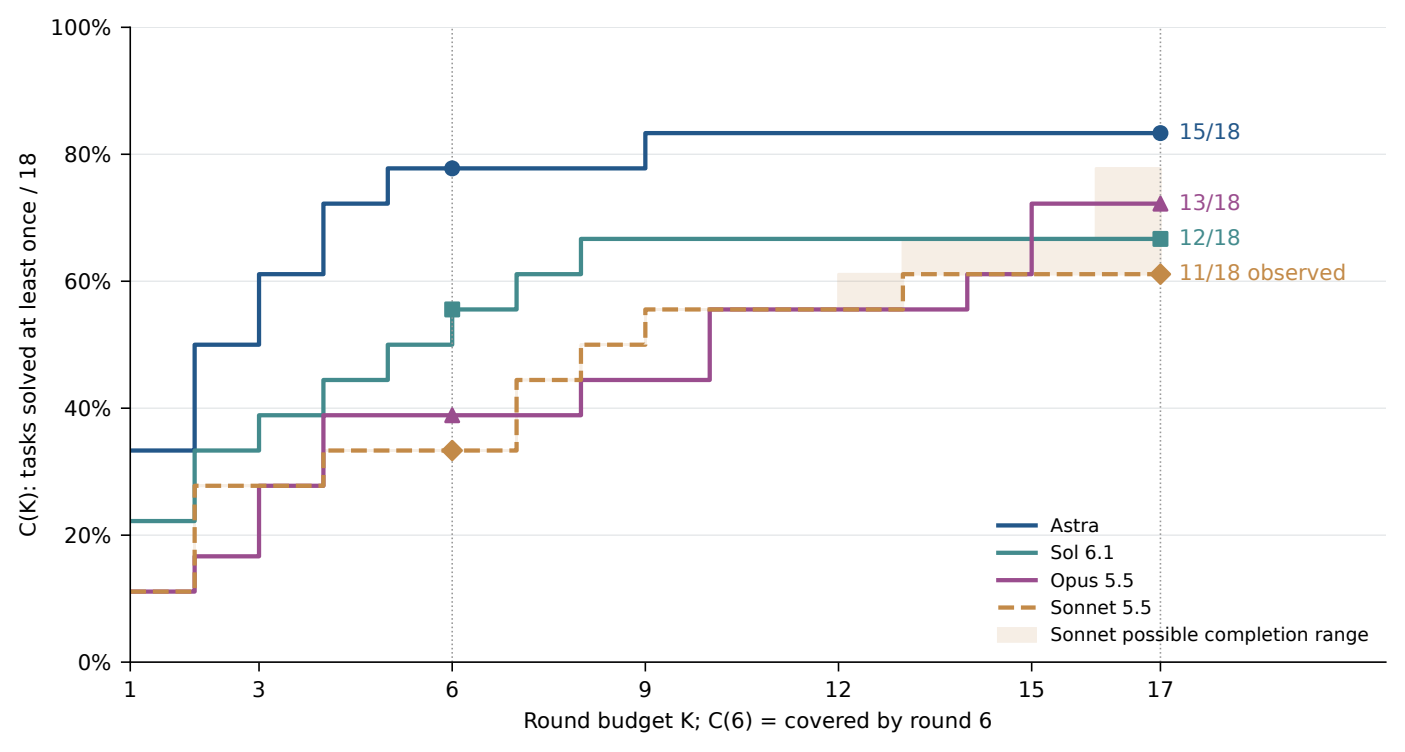
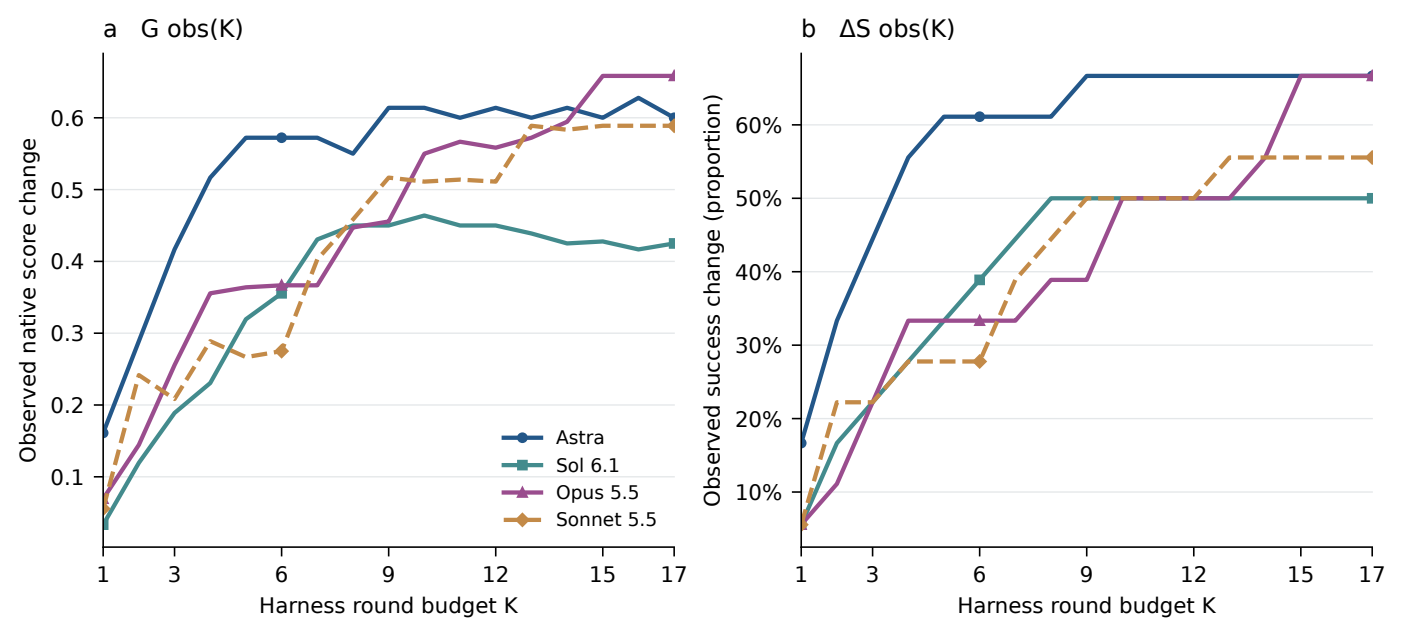

a Astra
b Sol 6.1
c Opus 5.5
d Sonnet 5.5 · 3 unfinished chains
arrange number
classify objects
cover blocks
deposit coin
fill pen holder
insert key
match & pick
organize table
pack objects
play Xylophone
plug in charger
pour
pour liquid → cup
push T
bottles → dustbin
stack blocks
store tools
swap blocks
0
1
3
6
9
12
15
17
Actual round k (blank = unobserved)
Native score q; ○ = native success
1.0
0.8
0.6
0.4
0.2
0.0

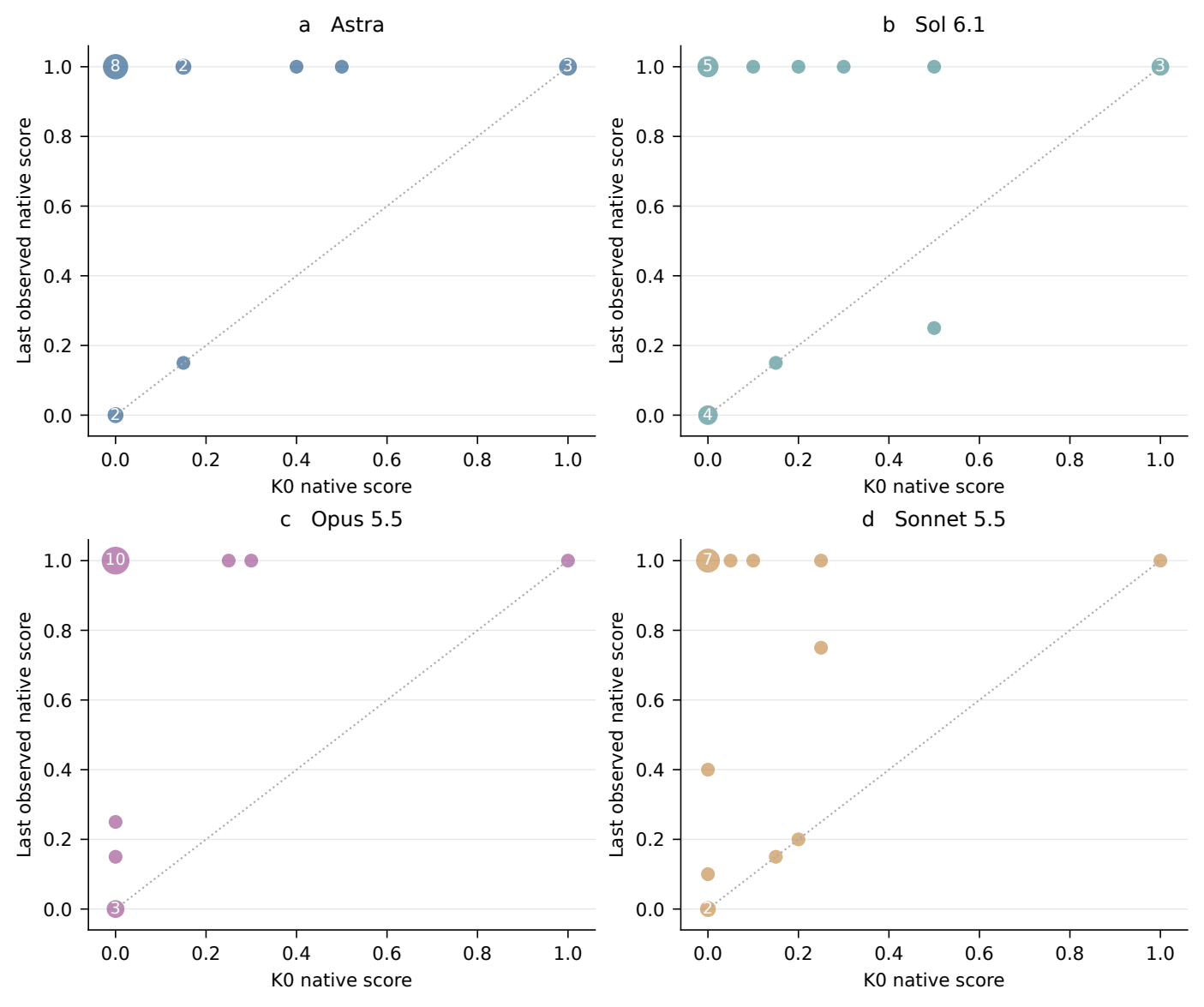

K0 (light) → best observed (solid); ranges are score ambiguity, not CI
Decoded stage fraction (11 sequential tasks)
100%
80%
60%
40%
20%
0%
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

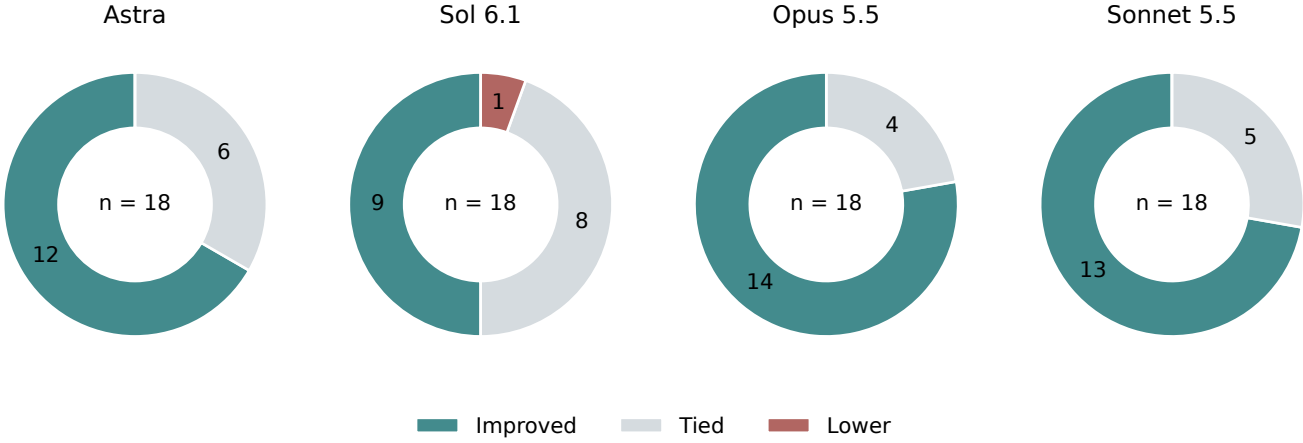

Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
arrange number
classify objects
cover blocks
deposit coin
fill pen holder
insert key
match & pick
organize table
pack objects
play Xylophone
plug in charger
pour
pour liquid → cup
push T
bottles → dustbin
stack blocks
store tools
swap blocks
1
3
6
9
12
15
17
NA
First Harness success T; × = unsolved, > = unfinished

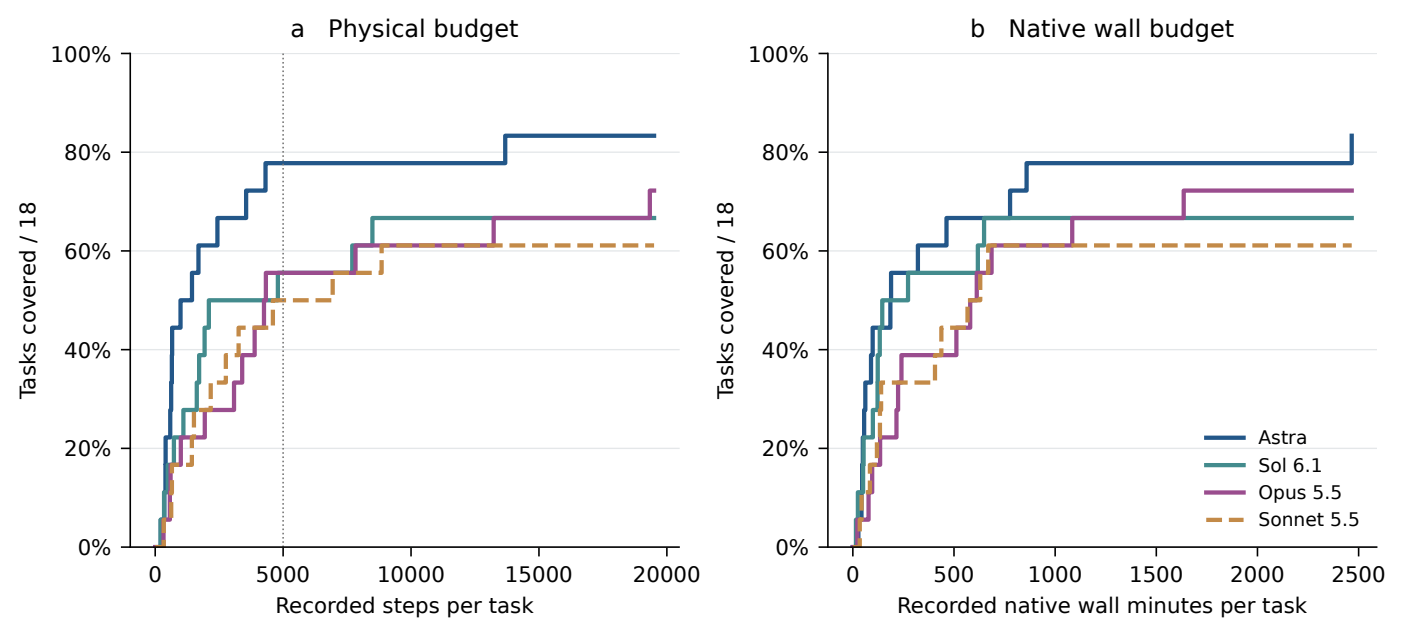

a K0 native success
40%
35%
30%
25%
20%
15%
10%
5%
0%
3/18
3/18
1/18
1/18
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
b K0 native score
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.242
0.264
0.086
0.111
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

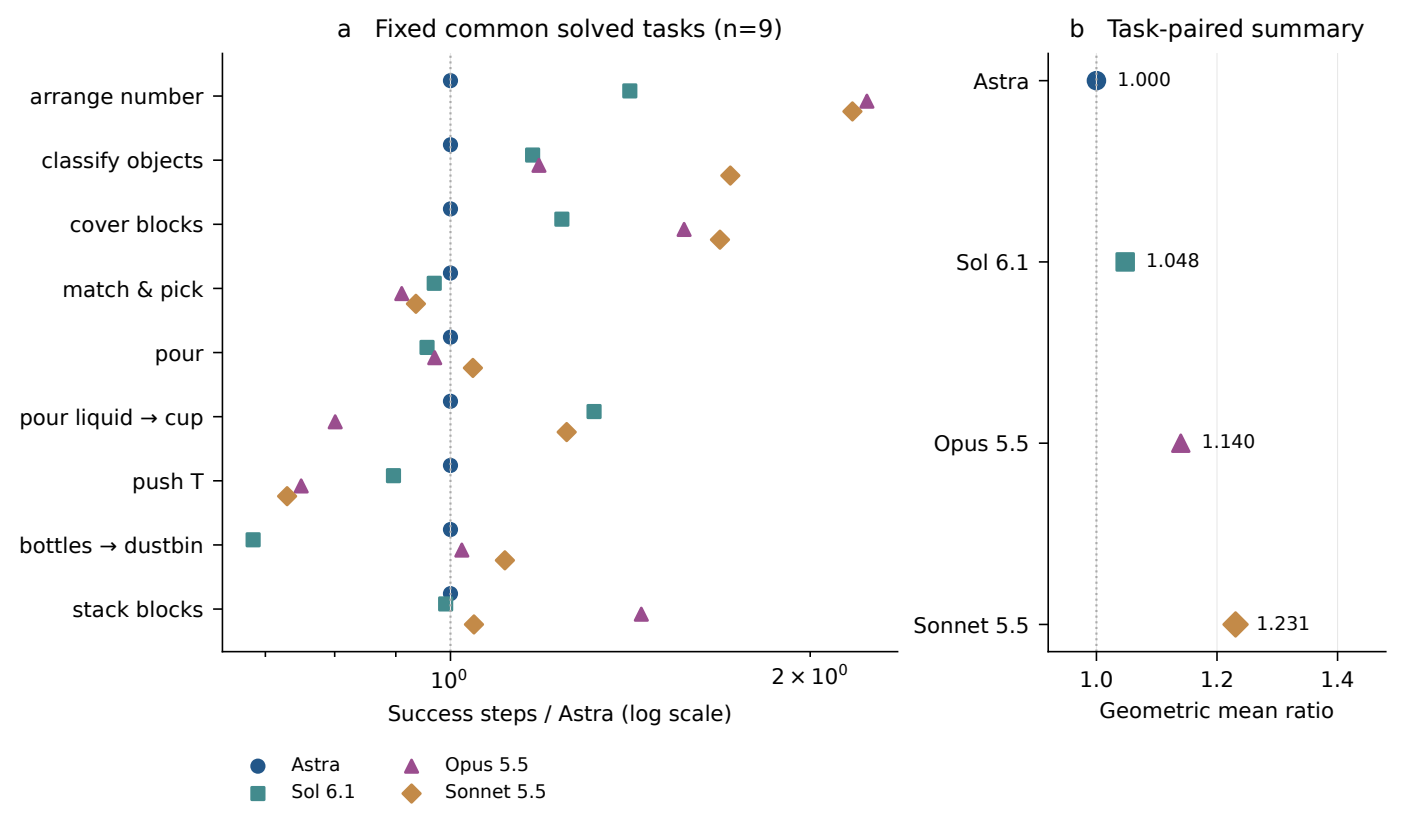

100%
80%
60%
40%
20%
0%
C(K): tasks solved at least once / 18
1
3
6
9
12
15
17
Round budget K; C(6) = covered by round 6
15/18
13/18
12/18
11/18 observed
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
Sonnet possible completion range

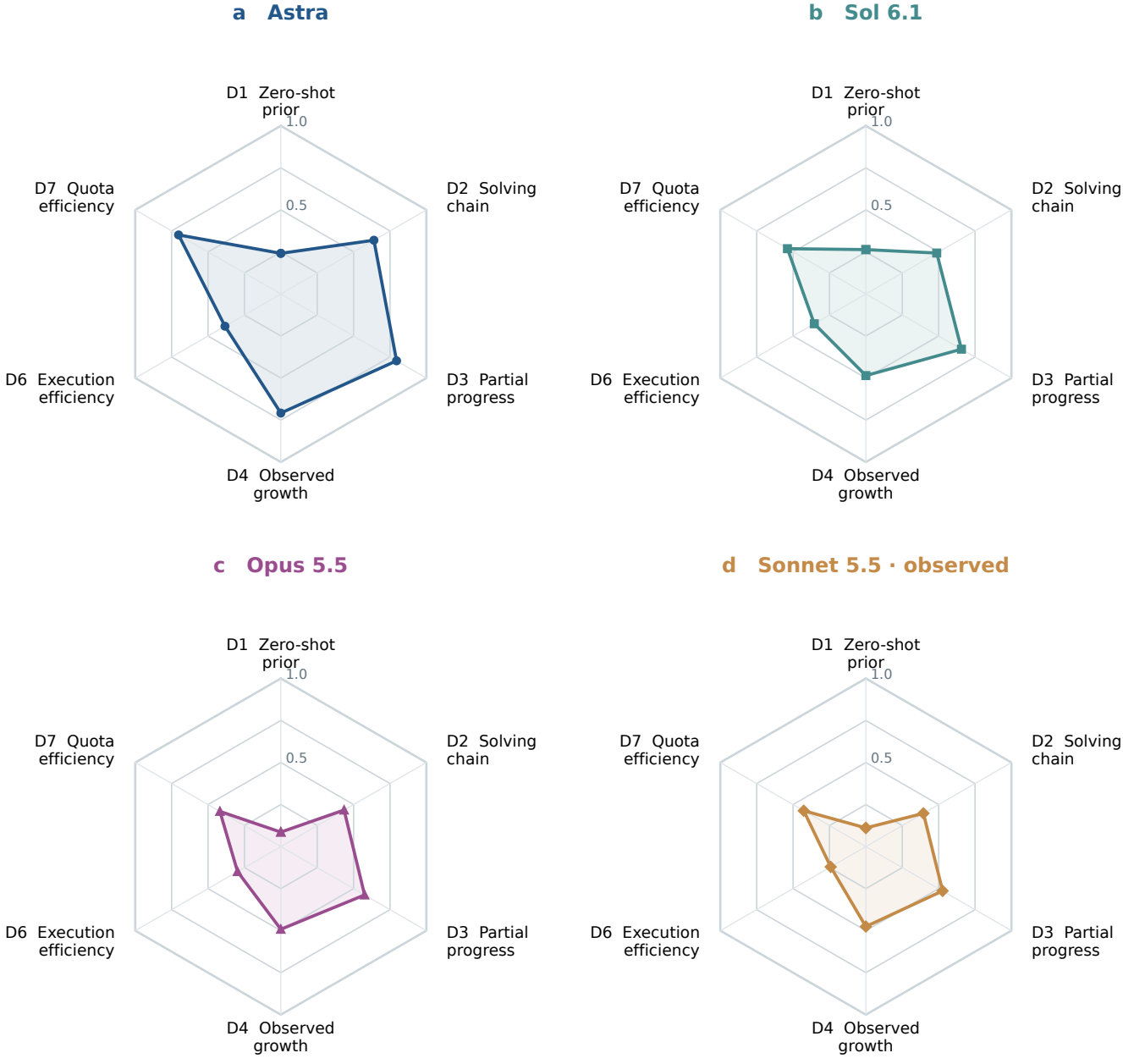

a G obs(K)
Observed native score change
0.1
0.2
0.3
0.4
0.5
0.6
1
3
6
9
12
15
17
Harness round budget K
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
b ΔS obs(K)
Observed success change (proportion)
10%
20%
30%
40%
50%
60%
1
3
6
9
12
15
17
Harness round budget K


a Astra
b Sol 6.1
c Opus 5.5
d Sonnet 5.5
Last observed native score
K0 native score
0.0
0.2
0.4
0.6
0.8
1.0
8
2
3
2
5
3
4
10
3
7
2

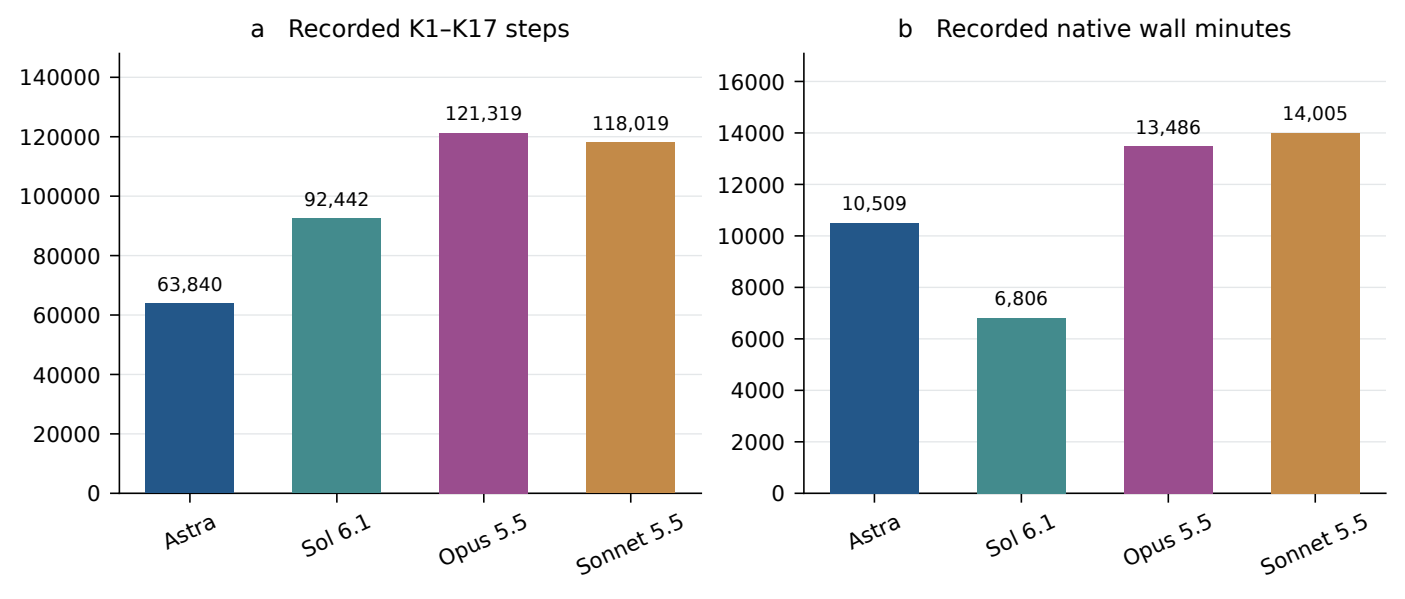

Astra
n = 18
12
6
Sol 6.1
n = 18
9
1
8
Opus 5.5
n = 18
14
4
Sonnet 5.5
n = 18
13
5
Improved
Tied
Lower

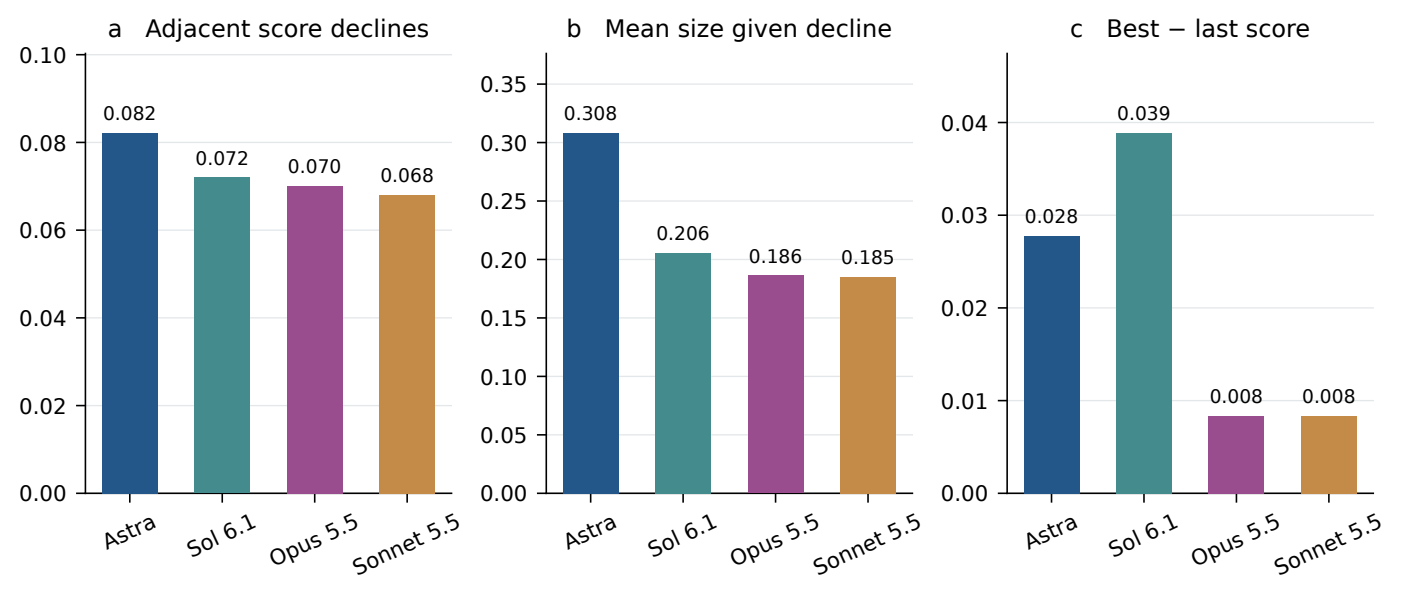

a Physical budget
b Native wall budget
100%
80%
60%
40%
20%
0%
Tasks covered / 18
0
5000
10000
15000
20000
Recorded steps per task
500
1000
1500
2000
2500
Recorded native wall minutes per task
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

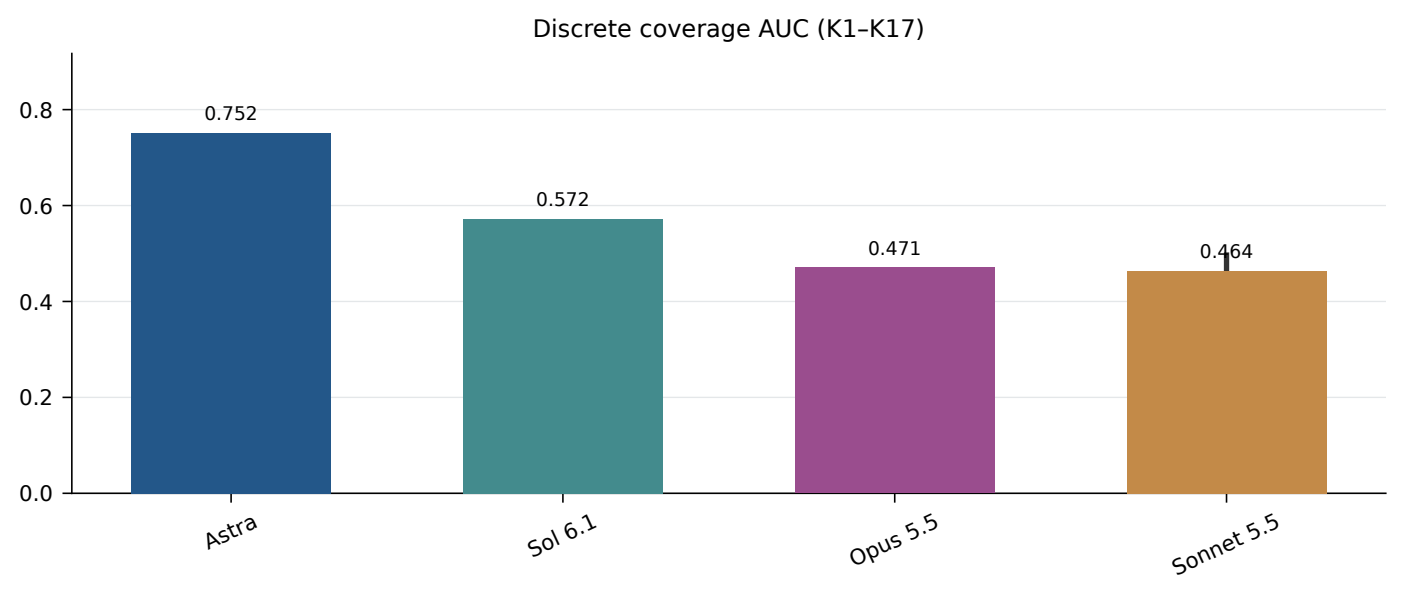

a Fixed common solved tasks (n=9)
arrange number
classify objects
cover blocks
match & pick
pour
pour liquid → cup
push T
bottles → dustbin
stack blocks
10^0
2×10^0
Success steps / Astra (log scale)
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
b Task-paired summary
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
1.000
1.048
1.140
1.231
1.0
1.2
1.4
Geometric mean ratio

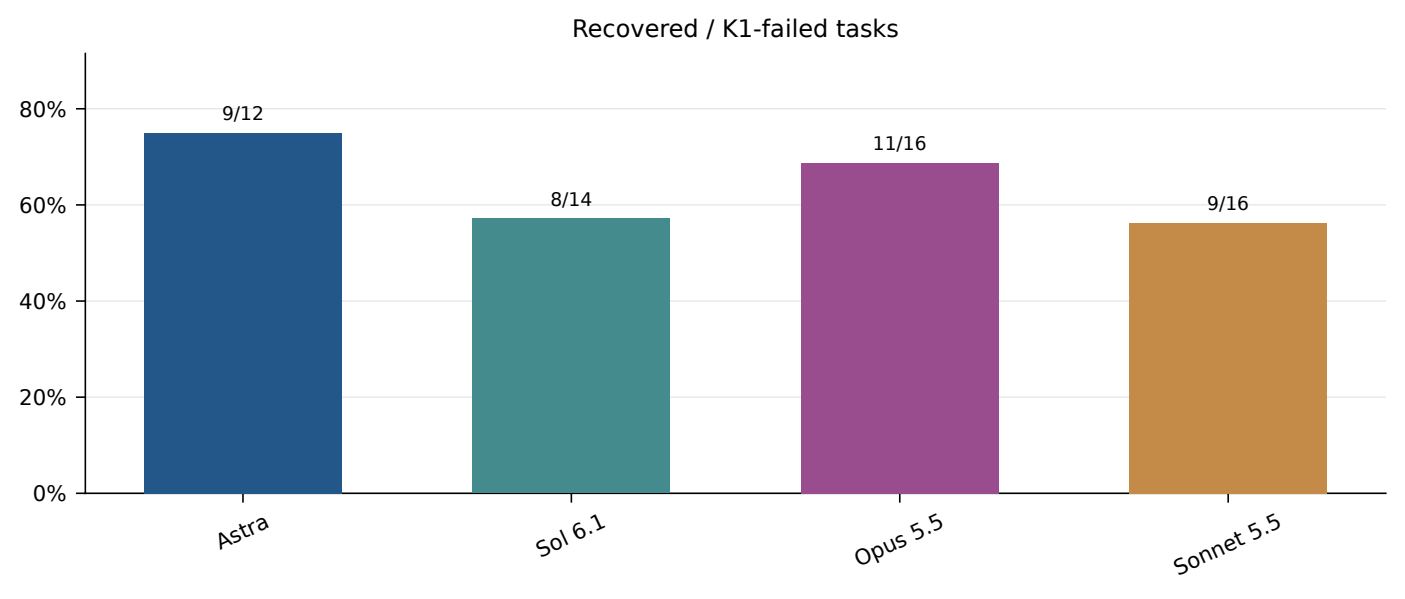


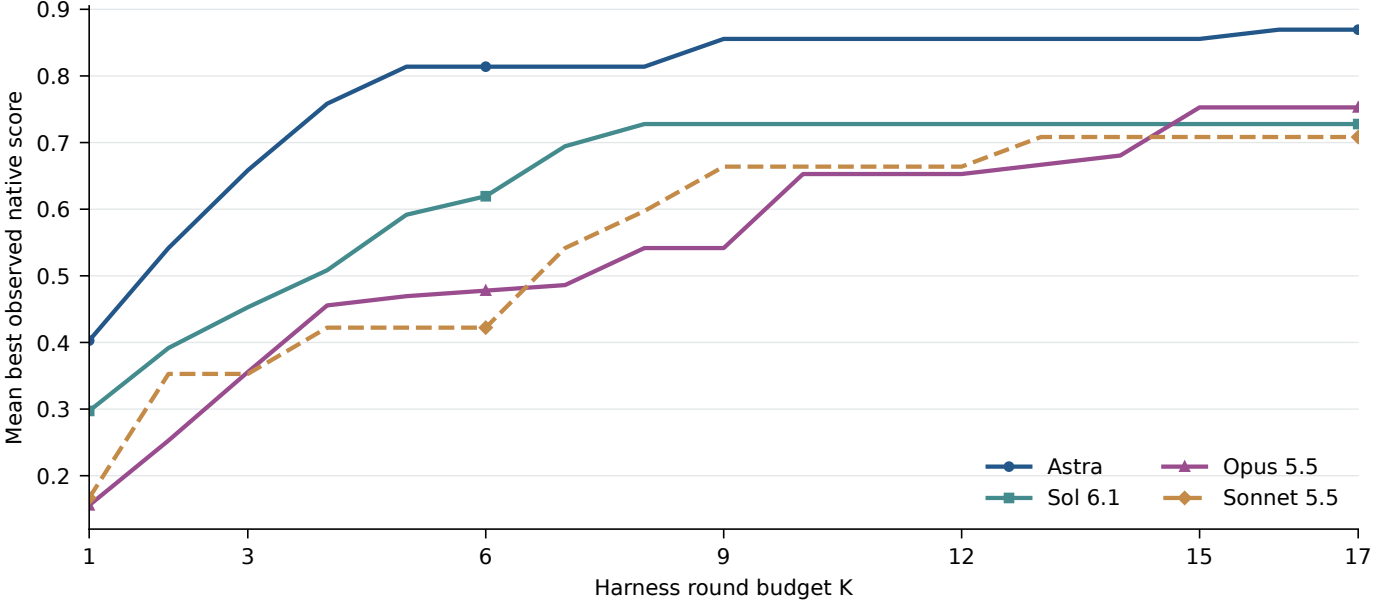
Fixed [0, 1] axes; higher = better by definition. D1/D2/D4/D7: 18 tasks; D3: 12 graded; D6: 9 fixed tasks.
Axes share evidence. Polygon size is not an independent capability measure. Sonnet remains incomplete.

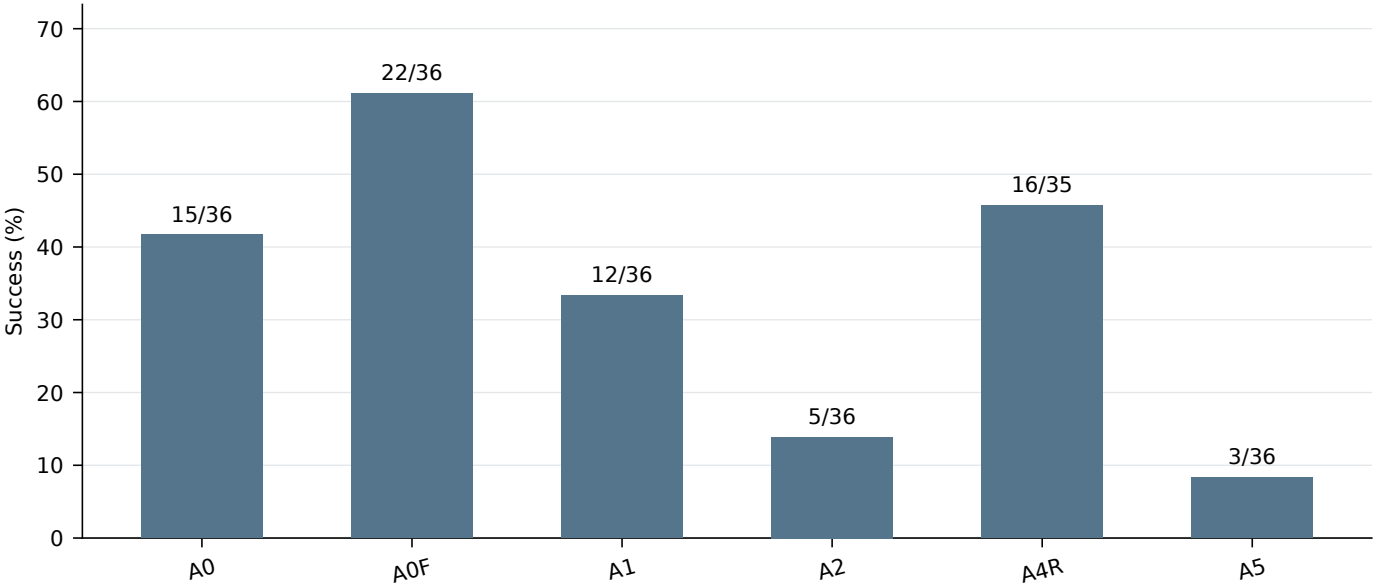

a Recorded K1–K17 steps
140000
120000
100000
80000
60000
40000
20000
0
63,840
92,442
121,319
118,019
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
b Recorded native wall minutes
16000
14000
12000
10000
8000
6000
4000
2000
0
10,509
6,806
13,486
14,005
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

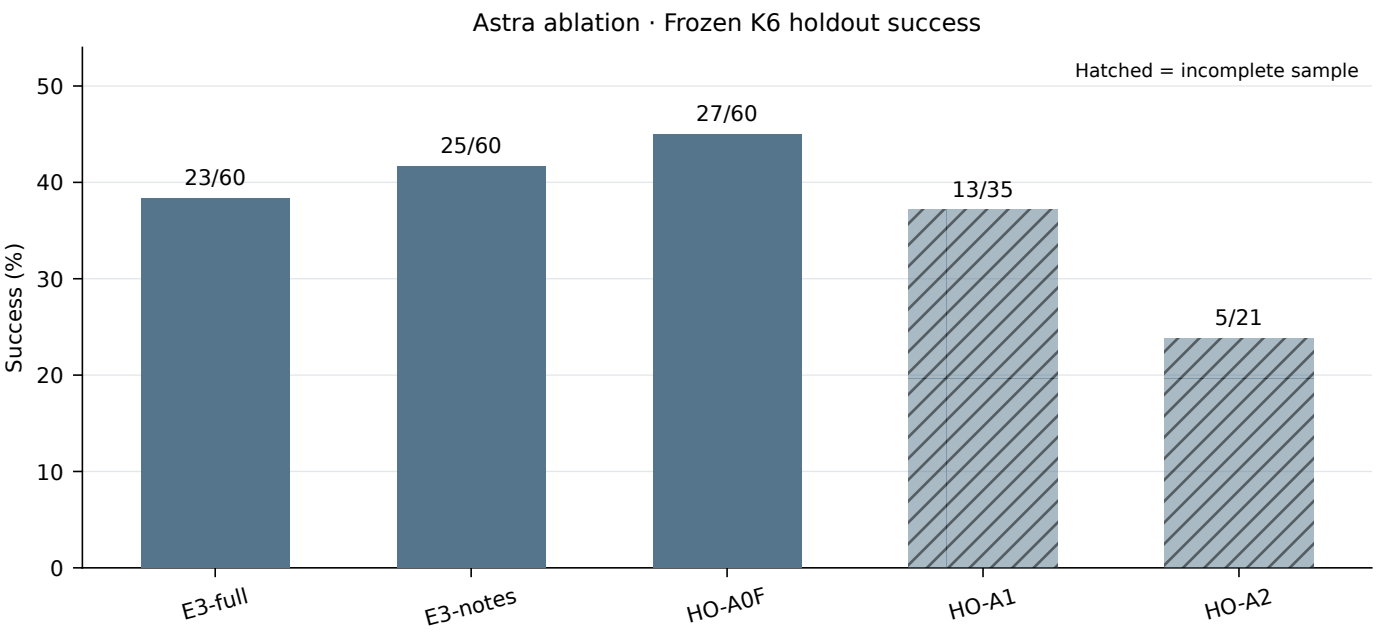

a Adjacent score declines
0.10
0.08
0.06
0.04
0.02
0.00
0.082
0.072
0.070
0.068
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
b Mean size given decline
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.308
0.206
0.186
0.185
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
c Best − last score
0.04
0.03
0.02
0.01
0.00
0.028
0.039
0.008
0.008
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

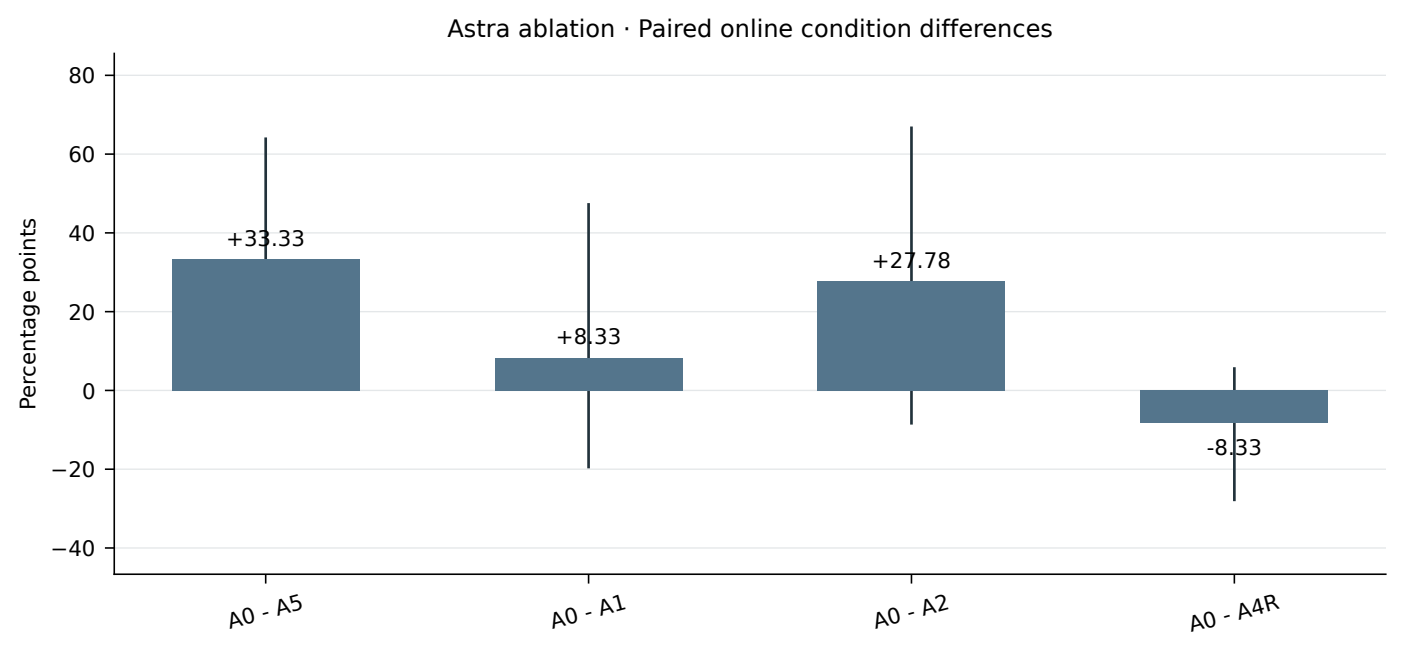

Discrete coverage AUC (K1–K17)
0.0
0.2
0.4
0.6
0.8
0.752
0.572
0.471
0.464
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

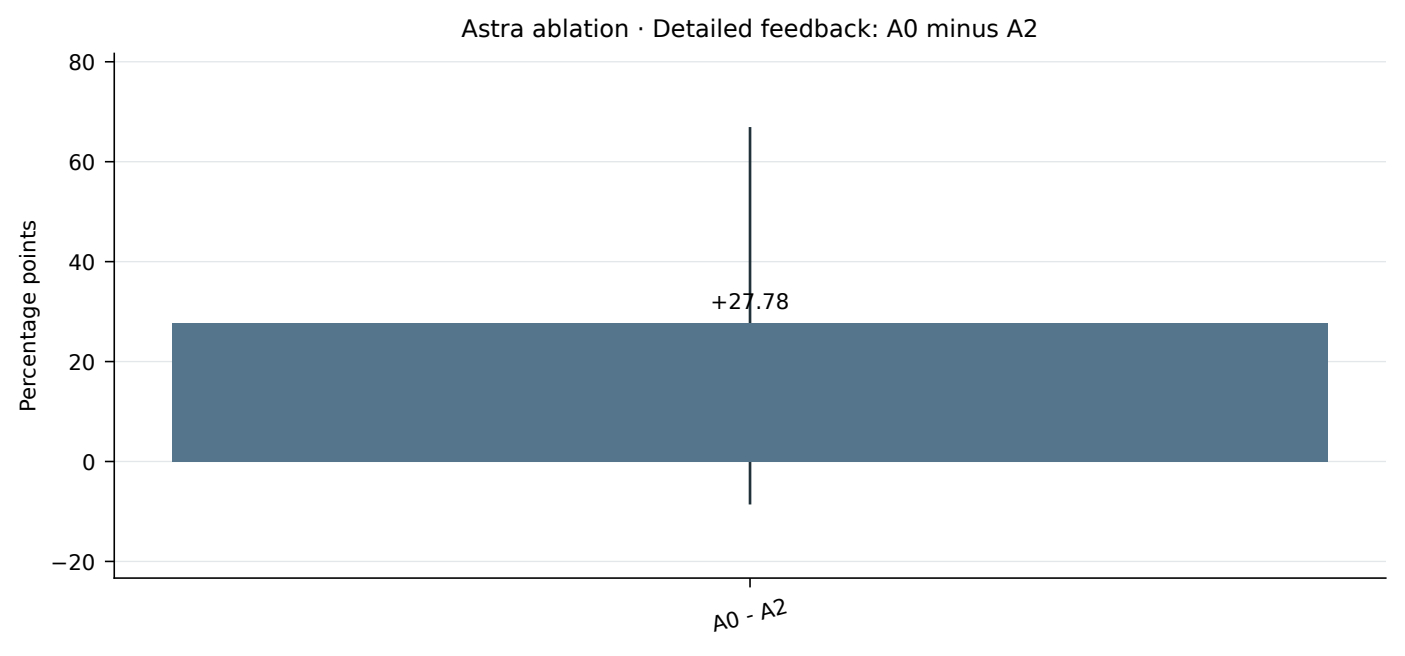

Recovered / K1-failed tasks
80%
60%
40%
20%
0%
9/12
8/14
11/16
9/16
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5

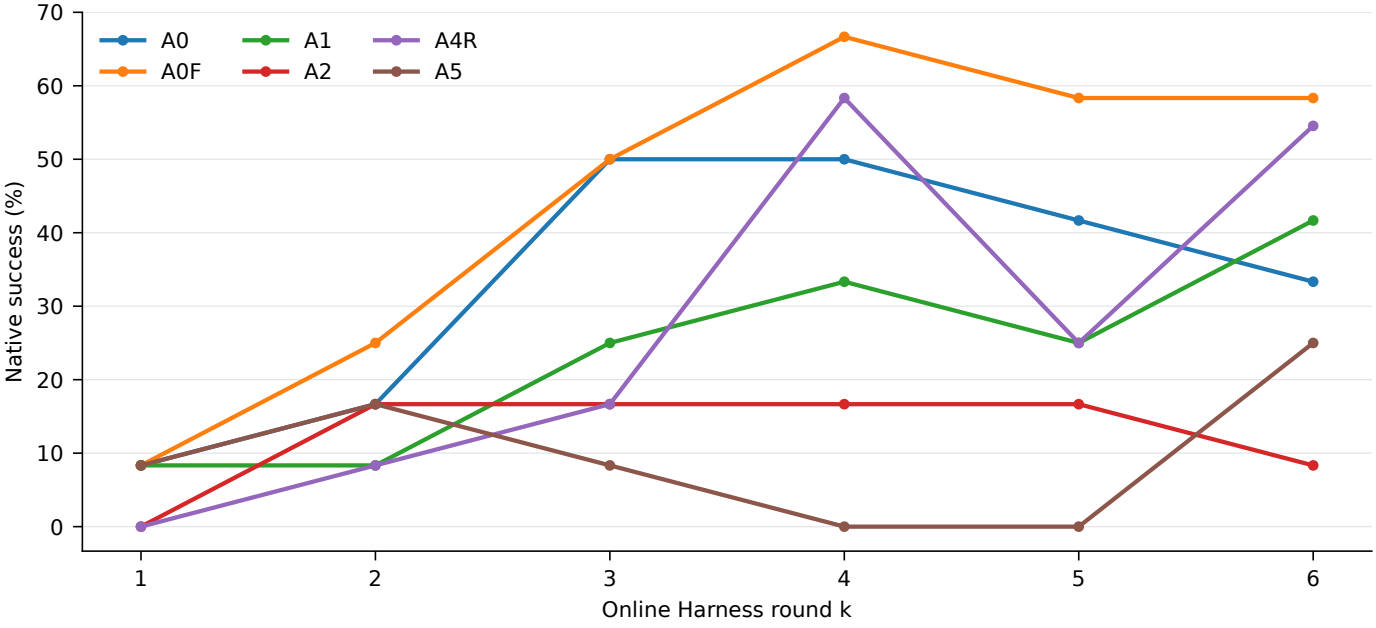

Mean best observed native score
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
1
3
6
9
12
15
17
Harness round budget K
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5


Astra ablation · Online K4–K6 success
Success (%)
0
10
20
30
40
50
60
70
15/36
22/36
12/36
5/36
16/35
3/36
A0
A0F
A1
A2
A4R
A5


Astra ablation · Frozen K6 holdout success
Hatched = incomplete sample
Success (%)
0
10
20
30
40
50
23/60
25/60
27/60
13/35
5/21
E3-full
E3-notes
HO-A0F
HO-A1
HO-A2


Astra ablation · Paired online condition differences
Percentage points
80
60
40
20
0
−20
−40
+33.33
+8.33
+27.78
-8.33
A0 - A5
A0 - A1
A0 - A2
A0 - A4R


Astra ablation · Detailed feedback: A0 minus A2
Percentage points
80
60
40
20
0
−20
+27.78
A0 - A2


A0
A0F
A1
A2
A4R
A5
Native success (%)
Online Harness round k
0
10
20
30
40
50
60
70
1
2
3
4
5
6